

Vertex-Minimal Paper Tori

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Abstract

A *paper torus* is an embedded polyhedral torus that is isometric to a flat torus in the intrinsic sense. We prove that there does not exist a paper torus with 7 vertices, and that there does exist a paper torus with 8 vertices.

1 Introduction

1.1 Context and Main Results

A *flat torus* is a quotient of the form $\mathbf{R}^2/\Lambda$, where Λ is a lattice of translations of $\mathbf{R}^2$. More concretely, a flat torus is what you get when you identify the opposite sides of a parallelogram by translations.

A *polyhedral torus* is a continuous, piecewise affine embedding $\phi : T \rightarrow \Omega \subset \mathbf{R}^3$. Here T is a flat torus that has been triangulated, and ϕ is a continuous embedding that is affine on each triangle of the triangulation. If, additionally, ϕ is an affine isometry on each triangle, we say that ϕ is *flat*. When ϕ is a flat embedding, ϕ tells how to build a torus in $\mathbf{R}^3$ out of finitely many triangles so that the cone angle around each vertex is 2π . We call a flat embedded polyhedral torus a *paper torus*. Some authors would call this an origami torus.

Surprisingly, paper tori exist. The 1960 paper of Y. Burago and V. Zalgaller [BZ1] gives the first constructions. Some of the simplest paper tori are called *diplo-tori*. These are described by U. Brehm [Br] in 1978, and also described in H. Segerman's book [Se, §6].

The 1995 paper [BZ2] proves that one can realize every isometry class of flat torus as a paper torus. The works of T. Tsuboi [T], and (independently) P. Arnoux,

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S. Lelièvre, and A. Málaga [ALM], show that every flat torus, except those made by identifying the sides of a rectangle, is realized by some diplotorus. (Both papers have a separate construction which works for the rectangular flat tori.) The 2024 preprint of F. Lazarus and F. Talerie [LT] gives a universal combinatorial type of triangulation which does the job simultaneously for all isometry types. Their triangulation 1217 vertices (and 2434 faces).

There are diplotori, based on a pair of regular pentagons, which have 10 vertices. In 2025, Vincent Tugayé [Tu], discovered a 9-vertex paper torus. At the same time, one needs at least 7 vertices to make a paper torus because there is no 6 vertex triangulation of a torus.

The 7-vertex triangulation of a torus, the *Moebius triangulation*, is unique up to combinatorial isomorphism. The Moebius triangulation gives the famous embedding of the complete graph K_7 in a torus. Figure 1.1 shows part of the universal cover of the Moebius torus, as well as a fundamental domain for it.

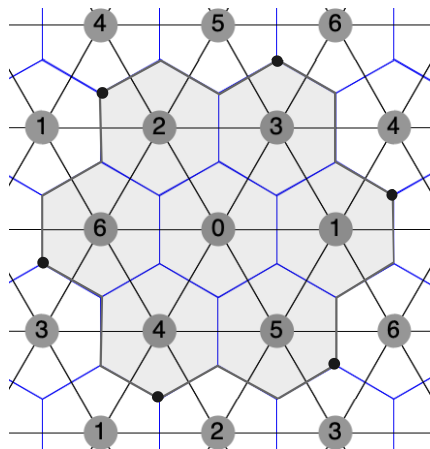


Figure 1.1: Part of the universal cover of the Moebius triangulation

In 1949, Á. Császár [Cs] showed that the Moebius torus has a polyhedral embedding. However, this embedding is not flat. See e.g. [G], [HLZ], and [L] for more detail and a discussion of related topics. The 2019 Ph.D. thesis of P. Quintanar Cortés, [QC] makes significant progress towards showing that there is no 7-vertex paper torus.

The results above leave open the question as to the minimum number of vertices needed to make a paper torus. The results in this paper settle the question.

Theorem 1.1 *There does not exist a 7-vertex paper torus.*

Theorem 1.2 *There exists an 8-vertex paper torus.*

1.2 Seven Vertices

The proof of Theorem 1.1 is actually rather easy. Here is the key step.

Lemma 1.3 (Hull) *Suppose Ω is an embedded 7-vertex polyhedral torus, not necessarily flat. Let H be the convex hull of Ω . Suppose Ω has all 7 vertices in ∂H . Then Ω has at least one vertex P such that all 6 triangles of Ω incident to P are in ∂H .*

I will deduce from the Hull Lemma that a paper torus must have a vertex P contained in the interior of the convex hull. I will then use Crofton's formula to show that the cone angle at P exceeds 2π . This contradiction finishes the proof of Theorem 1.1.

The Hull Lemma is a consequence of a stronger result contained in the 1991 paper [BE] of J. Bokowski and A. Eggert. Bokowski and Eggert use oriented matroids to classify the combinatorial types of 7-vertex polyhedral tori. One consequence of their classification is the *Hull Theorem*: there are no embedded 7-vertex polyhedral tori at all having all 7 vertices on their convex hull boundary. (See the remarks after [BE, Theorem 3.7]. This is my name for their result.) In my opinion, had these authors asked the flatness question for 7-vertex polyhedral tori, they most likely would have been able to answer it.

In order to keep this paper self-contained, I will give a light and entirely combinatorial computer-assisted proof of the Hull Lemma. It is not very hard to promote the Hull Lemma to the Hull Theorem, but since this is not needed for the proof of Theorem 1.1 I will not do it. I give the argument in my supplementary notes [S].

1.3 Eight Vertices

Motivated by our proof of Theorem 1.1, I tried to rule out the existence of 8-vertex paper tori having all 8 vertices on their convex hull boundary. This did not work, so I decided to find these kinds of examples and study them.

There are 7 combinatorial types of triangulation of the torus having 8 vertices. Of these 7 there is (in my opinion) a *worst triangulation* and a *best triangulation*. The worst triangulation is obtained from the Moebius triangulation by subdividing one of the triangles. The best triangulation has degree sequence 66666666 and is vertex-transitive. This triangulation is very much like the Moebius triangulation. Figure 1.2 below shows part of the universal cover of this triangulation, as well as a fundamental domain. I will explain the meaning of the blue triangles below. The triangles all have the form $(a, a + 1, a + 3)$ and $(a, a + 2, a + 3)$ with indices taken mod 8.

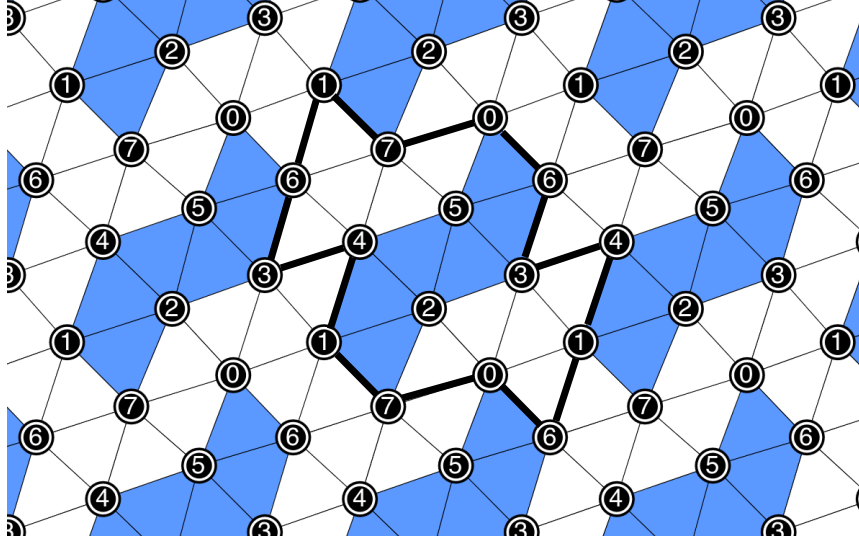


Figure 1.2: The universal cover of the best 8 vertex triangulation

After an intense period of experimentation, I discovered some 8-vertex paper tori. In §4 I will explain in detail how I found these. In the example below all the coordinates except z_0, z_1, z_2 are exact. The listings of z_0, z_1, z_2 are 32-digit truncations of the true values.

$$\begin{array}{rcll}
 +0.64 & -0.20 & 1 & \\
 -1.09 & +0.38 & z_1 & \\
 -0.25 & +0.51 & z_2 & z_1 = 0.0206\ 6632\ 6669\ 8443\ 6159\ 8992\ 3371\ 8861 \\
 +0.78 & +0.62 & z_3 & z_2 = 0.0048\ 5312\ 7706\ 5192\ 8720\ 4090\ 7479\ 6169 \\
 -0.78 & -0.62 & z_3 & z_3 = 0.0082\ 2752\ 1455\ 6137\ 1645\ 5791\ 2547\ 8661 \\
 +0.25 & -0.51 & z_2 & \\
 +1.09 & -0.38 & z_1 & \\
 -0.64 & +0.20 & 1 &
 \end{array} \quad (1)$$

Following a suggestion of Peter Doyle, I call this example and ones like it *pup tents*, on account of their appearance. See Figure 4.1. Specifically, a pup tent is an 8-vertex paper torus with the following properties:

- It is based on the 8-vertex uniform triangulation.
- It has order 2 symmetry given by the map $(x, y, z) \rightarrow (-x, -y, z)$.
- It has 6 faces in its convex hull boundary, corresponding to the blue faces in Figure 1.2.

Here I discuss the proof that the example in Equation 1 is really the truncation of an 8-vertex paper torus. If we leave z_1, z_2, z_3 unspecified, the example in Equation 1 is part of a 3-parameter family of examples we call $T(z_1, z_2, z_3)$. Let $\theta_k(z_1, z_2, z_3)$ be the cone angle of $T(z_1, z_2, z_3)$ at vertex k . Let

$$F(z_1, z_2, z_3) = (\theta_1, \theta_2, \theta_3). \quad (2)$$

If $F(z_1, z_2, z_3) = (2\pi, 2\pi, 2\pi)$ and $T(z_1, z_2, z_3)$ is embedded then we have our proof. Let dF be the Jacobian of F , computed at our example. Here we show dF with 4 digits of accuracy.

$$\begin{bmatrix} -1.6627\dots & 1.4699\dots & -0.0244\dots \\ 1.4699\dots & 0.04714\dots & -1.1300\dots \\ -0.0244\dots & -1.1300\dots & 1.9196\dots \end{bmatrix} \quad (3)$$

The matrix dF is an invertible matrix, and all of its eigenvalues exceed $3/10$ in absolute value. This means that the inverse dF^{-1} is also small. In other words, dF is a well-conditioned matrix.

Our proof is an interplay of three things. First, we show that our example is robustly embedded in the sense that moderate perturbations keep it embedded. Second, as we say above, our example is very close to being flat. Third we show, by bounding derivatives, that our Jacobian matrix stays fairly constant and well conditioned throughout a healthy neighborhood of our example. We then use an effective version of the Inverse Function Theorem so show that a true paper torus is quite near our example. We will add enough slack to our proof so that it would work for examples which are fairly near the one in Equation 1.

Given the flexibility of our argument, we get the following easy corollary.

Corollary 1.4 *The space $\mathcal{X}$ of pup tents contains an open 6-dimensional ball.*

Proof: We start with the example based on Equation 1 and we then perturb our points symmetrically by changing the X and Y coordinates but not the Z -coordinates. As long as the perturbation is small enough, the Inverse Function Theorem selects Z -coordinates for us which give a pup tent. We have 8 free parameters to play with and then these are grouped into a 2-dimensional foliation under the action of similarities of the XY -plane which fix the origin. A cross-section of this foliation gives us our 6-dimensional open manifold of inequivalent pup tents. ♠

1.4 Further Developments

Since I wrote the first version of this paper, there have been a number of developments. First of all, some of us at an ICERM workshop in August 2025 printed out pup tents from a paper model I made (with advice from Alba Málaga). With some amount of dexterity, you can use the paper model in [S] to make your own. This model is based on a pup tent that has slightly different coordinates than the one in Equation 1, but the shape is so close you can hardly notice the difference.

Second, Peter Doyle made an amazing discovery of some degenerate versions of the pup tent which realize every flat structure. We call the set of these degenerate pup tents *the golden valley*. Peter and I then used the golden valley to get some further results about 8 vertex paper tori. Our sequel paper [DS] has many of our findings. Here are two example results:

1. Any flat torus which lacks reflection symmetry – this accounts for almost all of them – is realized by a pup tent.
2. Given any $\epsilon > 0$ there is an 8-vertex paper torus within ϵ of the unit equilateral triangle in the Hausdorff metric.

Also, the proofs in [DS] give a second and independent proof of Theorem 1.2.

Third, after hearing about the pup tent, my undergraduate student, Zhengyu (Byron) Zou tried for similar results in hyperbolic geometry. Recently he found a 10-vertex intrinsically hyperbolic embedding of a genus 2 surface in $\mathbf{H}^3$. This is vertex-minimal, because you need 10 vertices to triangulate a genus 2 surface. See [Z]. Even more recently, he discovered a 12-vertex hyperbolic embedding of a genus 2 surface which has order 2 symmetry. The order 2 symmetry implements the hyperelliptic involution.

1.5 Computer Assistance

My proof is heavily computer assisted, though all the important calculations are done with exact integer arithmetic. The reader can download all the code for this paper at www.math.brown.edu/~res/Papers/VertexMinimal.tar.

I used ChatGPT-4o and ChatGPT-5 in various ways to help me with this project. Since this is my first time doing a project with the assistance of ChatGPT, I think that this warrants some discussion. I used ChatGPT to discuss high level mathematical ideas, to find typos and other inconsistencies in my paper, to format data both for the paper and for my computer code, and also to document the efficiency

of my programs. There was one astounding thing ChatGPT5 did. In a stroke of brilliance, it found on its own the expression for the denominator of g_{11} at the end of the paper. I might not have arrived at that. I did not let ChatGPT compose prose for me and I ignored its writing advice. I want to retain 100 percent of my own voice when I write something. Overall, I found ChatGPT to be a helpful and enthusiastic companion as I worked.

1.6 Acknowledgements

I thank Samuel Lelièvre and Alba Málaga for telling me all about flat tori (over a period of some years) and in particular telling me about the flatness problem and supplying me with some historical context. The main thing that inspired me to work on this project was the infectious enthusiasm of Samuel and Alba. I also thank Peter Doyle, Fabian Lander, Noah Montgomery, Stepan Paul, Saul Schleimer, and Byron Zou for helpful discussions about this material. I thank Frank Lutz for making his manifolds webpage, which lets you download the combinatorial data for triangulations of surfaces and other manifolds. This page was very useful to me.

I thank the IHES, the Hamilton Institute, and the Isaac Newton Institute, where I started working on this paper. This work is also supported by a Simons Sabbatical Fellowship and the Mazur Chair at IHES.

2 Proof of Theorem 1.1

2.1 Crofton's Formula

Crofton's Formula is a classic result from integral geometry. See [S]. It applies to any rectifiable arc on the unit sphere S^2 . We just need this result for finite unions of arcs of great circles, which we call *spherical polygons*.

The space of oriented great circles in S^2 is canonically bijective with S^2 itself, and inherits a canonical probability measure σ . Given a spherical polygon γ and a great circle C , we let $\#(C \cap \gamma)$ denote the number of intersection points. (We can ignore the finitely many great circles for which this is infinite.) Crofton's formula is as follows.

$$\text{length}(\gamma) = \pi \int_S \#(C \cap \gamma) d\sigma. \quad (4)$$

At least for spherical polygons, this has a swift proof. Both sides of Equation 4 are additive, so it suffices to prove the equation for a spherical polygon that is just an arc of a single great circle. By continuity, it suffices to prove the result for great circle arcs whose length is a rational multiple of 2π . By additivity once again, it suffices to prove Crofton's formula for the great circle itself. But then the formula is obvious.

Here is a well-known consequence of Crofton's formula.

Lemma 2.1 *Suppose γ is a spherical polygon which is also a topological loop. If γ has length at most 2π then γ lies in some hemisphere of S^2 .*

Proof: Crofton's formula says that some great circle C intersects γ at most once. But then γ lies in one of the two hemispheres defined by C . When γ has length exactly 2π we can shorten γ a bit by cutting a small corner off. The shortened curve then lies in a hemisphere. Taking a limit as the cut corner tends to 0 in length, we see that γ itself lies in a hemisphere. ♠

2.2 Proof of the Main Theorem

Now suppose that Ω is a flat embedded 7-vertex polyhedral torus. Let H be the convex hull of Ω . Call a vertex of Ω *interior* if it does not lie in ∂H . Given any vertex P of Ω let Ω_P denote the union of 6 triangles of Ω incident to P .

Lemma 2.2 *Ω has an interior vertex.*

Proof: Suppose that Ω does not have an interior vertex. By the Hull Lemma, Ω has a vertex P such that all 6 triangles of Ω incident to P lie in ∂H . Let Ω_P be the union of triangles incident to P as above.

A *support plane* for H is a plane which intersects the boundary ∂H and is disjoint from the interior of H . Since the cone angle of Ω at P is 2π , and since $\Omega_P \subset \partial H$, we see that H cannot be strictly convex at P . Hence some support plane through P intersects Ω_P in (at least) a line segment through P that contains P in its relative interior. But then the intersection of ∂H with a small sphere centered at P is a spherical polygon consisting of two arcs which connect the same pair of antipodal points.

Since the total length of γ is 2π , and since a great semicircle is the unique path on S^2 of length at most π connecting two antipodal points, we see that γ must be a union of two great semicircles. Hence Ω_P is contained in the union of two planes whose intersection contains P .

One of these two planes, Π , contains at least 3 consecutive triangles of Ω_P . But then Π contains at least 5 vertices of Ω . Since the 1-skeleton of the triangulation of Ω is the complete graph K_7 , we see that Π contains an embedded copy of K_5 . This contradicts the fact that K_5 is not planar. ♠

Now we know that our flat embedded 7-vertex torus Ω has an interior vertex P .

Lemma 2.3 *The cone angle θ at P exceeds 2π .*

Proof: We translate so that P is the origin. Let S^2 be the unit sphere. Let θ_P denote the cone angle of Ω at P . Let $\ell(\cdot)$ stand for length.

Let Ω_P be the union of 6 triangles of Ω incident to P , as above. Let $\widehat{\Omega}_P$ denote the union of rays emanating from P whose initial portions are contained in Ω_P . Let $L_P = \widehat{\Omega}_P \cap S^2$. We have $\theta_P = \ell(L_P)$.

Since P is in the interior of the convex hull of Ω and since $\widehat{\Omega}_P$ contains all the vertices of this convex hull, we see that $\widehat{\Omega}_P$ cannot lie in any halfspace bounded by a plane through the origin. This means that L_P cannot lie in any hemisphere of S^2 . By Lemma 2.1, we have $\ell(L_P) > 2\pi$. Hence $\theta_P > 2\pi$. ♠

This completes the proof of the Main Theorem modulo the Hull Lemma. We mention again that the Hull Lemma is an immediate consequence of (what I call) the Hull Theorem in [BE]. Readers who are keen to see the existence of the 8-vertex paper torus can stop reading now. Readers who want a self-contained proof of the Hull Lemma should keep reading.

2.3 Proof of the Hull Lemma

We suppose that Ω is an embedded 7-vertex polyhedral torus having 7 points on its convex hull H . We can perturb so that the points are in general position. This makes H into a triangulated solid polyhedron with 7 vertices, 15 edges, and 10 faces. One property we will use repeatedly is that Ω is *neighborly*: Every two vertices of Ω are joined by an edge in the triangulation. In particular, every edge of H is an edge of the triangulation of Ω .

We call the 15 edges in ∂H the *external edges*. We call the remaining 6 edges *internal edges*. We say that an *internal edge pattern* is a choice of 6 distinguished edges from the 1-skeleton of the triangulation, normalized (by symmetry) so that the first internal edge is (01). There are $\binom{20}{5} = 15504$ different internal edge patterns.

We say that an *external triangle* of Ω is one that lies in ∂H , and an *internal triangle* is one that does not. Each internal edge is incident to two internal triangles.

Lemma 2.4 *An internal triangle cannot be bounded by 3 external edges.*

Proof: Suppose that such a triangle exists. Call it τ . If all three edges of τ lie in ∂H and $\tau \not\subset \partial H$ then τ separates H into two components, both of which contain vertices of Ω in their interior. Any path in H connecting two such vertices must intersect τ . On the other hand $\Omega - \tau$ is path connected. This is a contradiction. ♠

By Lemma 2.4, the internal edge pattern determines the set of internal triangles and the set of external triangles. It is worth pointing out a triangle of ∂H is not necessarily a triangle of Ω .

Since Ω is neighborly, the degree of a vertex in the triangulation of ∂H is just the number of external edges incident to that vertex, and this is the same as 6 minus the number of internal edges incident to the vertex. In particular, since H is a convex polyhedron and ∂H is triangulated, the degree of each vertex in the triangulation of ∂H is at least 3. In other words, at most 3 internal edges are incident to the same vertex of Ω . We eliminate all internal edge patterns having fewer than 3 internal edges incident to some vertex.

Now we consider the remaining internal edge patterns. For each vertex (q) we have the list $\{(q, v_i) \mid i = 1, \dots, K_q\}$ of K_q external edges incident to q . We write $K = K_q$ and order these $K = K_q$ vertices cyclically according to the link of (q) in Ω . The link of L_q of (q) in ∂H is some permutation $(w_1, \dots, w_K)$ of $(v_1, \dots, v_K)$.

The following lemma lets us deduce the cycle structure on H from the pattern of internal edges.

Lemma 2.5 (Cycle Rule) *In order to be a viable candidate for the link of (q) in ∂H , the link $(w_1, \dots, w_K)$ must satisfy two properties.*

1. (w_i, w_{i+1}) must be an external edge for all i . Indices are taken cyclically
2. The cycle $(w_1, \dots, w_K)$ must be a dihedral permutation of $(v_1, \dots, v_K)$.

Proof: The necessity of Condition 1 is obvious. Condition 2 requires some explanation. Let P be vertex (q) . Consider the picture in ∂H at P . Since the points are in general position, ∂H is a proper convex cone near P . We let Π be a plane parallel to a support plane through (q) that just cuts off a small corner of H near (q) . Consider the intersection $\partial H \cap \Pi$. This is a convex K -gon Δ , and the cyclic order of Δ is given by $(w_1, \dots, w_K)$.

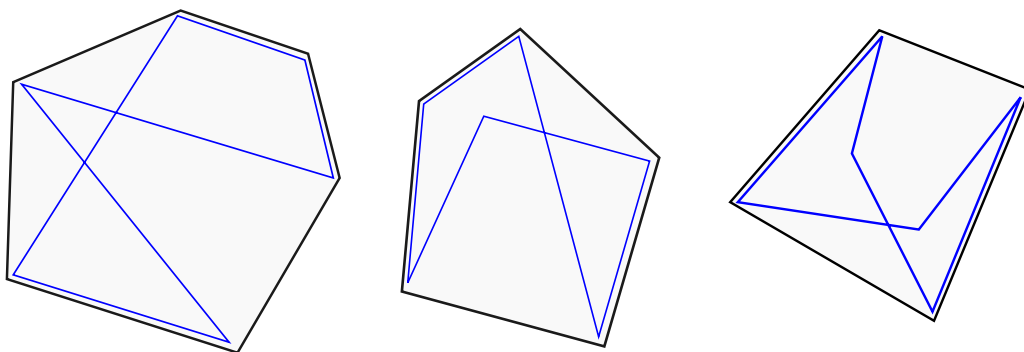


Figure 2.1: Δ (black) and γ (blue)

At the same time, and using the notation from the proof of Lemma 2.3, let $\gamma = \hat{\Omega}_P \cap \Pi$. The polygonal loop γ is contained in the region bounded by Δ and visits the vertices of Δ in the order $(v_1, \dots, v_K)$. Figure 2.1 shows some examples, with γ slightly moved off Δ to make the drawing more clear. Note that if $K < 6$ then there will be some extra vertices to γ as well; this does not matter in the argument. If the permutation is not dihedral then γ cannot be an embedded loop. But then Ω_P is not embedded, a contradiction. ♠

Using the Cycle Rule (and our computer code) we eliminate all remaining internal edge patterns except 6. These 6 are the same up to combinatorial isomorphism. Figure 2.2 shows one of them. Evidently, all 6 triangles incident to vertex (2) are external triangles. This completes the proof of the Hull Lemma.

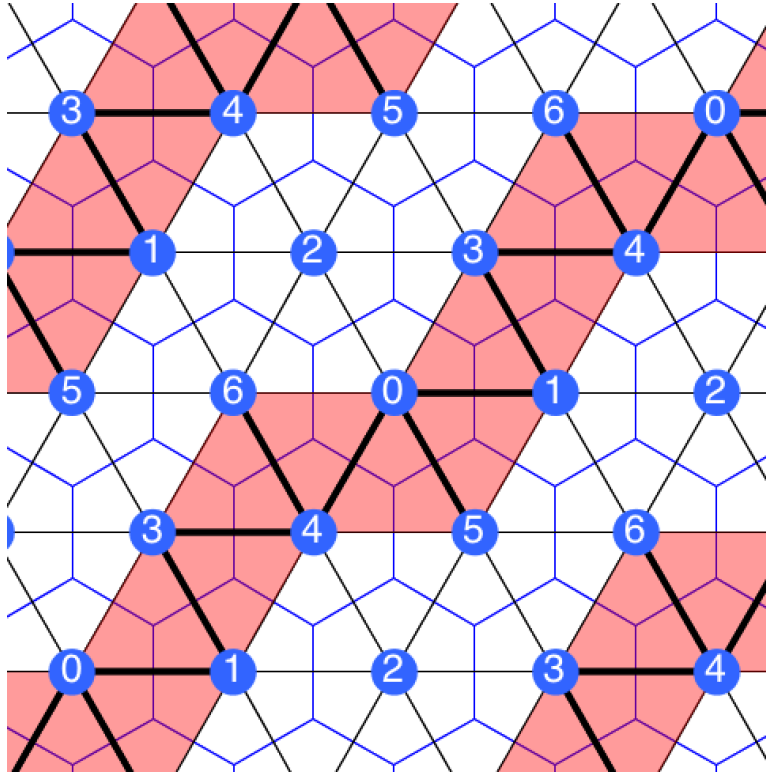


Figure 2.2: The one remaining pattern

Remark: As mentioned in the introduction, we have a separate geometric argument that rules out the 6 patterns like the one in Figure 2.2, thereby giving a second proof of the Hull Theorem. See my supplementary notes [S] for this argument.

3 Proof of Theorem 1.2

Throughout this chapter $\|\cdot\|$ denotes the Euclidean norm and $\|\cdot\|_\infty$ denotes the maximum of the absolute values of vector or matrix entries. Also, $\angle$ denotes angles.

3.1 Robust Embedding

Let $T = T(z_1, z_2, z_3)$ be the example from Equation 1. Let $T(z'_1, z'_2, z'_3)$ be as in Equation 1, but with variables z'_1, z'_2, z'_3 . Let

$$|T' - T| = \max(|z'_1 - z_1|, |z'_2 - z_2|, |z'_3 - z_3|). \quad (5)$$

We say that T is *K-robustly embedded* if and only if T' is embedded whenever $|T' - T| \leq K$. In this section we prove the following result.

Lemma 3.1 (Robust Embedding) *T is 10^{-4} robustly embedded.*

We have $120 = 24 + 72 + 24$ triangle pairs to consider. The first 24 are pairs having 0 vertices in common. The next 72 are pairs having 1 vertex in common. The last 24 are pairs having 2 vertices in common. We don't need to check these last 24.

Lemma 3.2 *Let T' be a perturbation of T such that all triangle pairs with no vertices in common are disjoint and all pairs with one vertex in common only intersect at the common vertex. Then T' is embedded.*

Proof: We will argue by contradiction. If T' is not embedded, then we can find a pair of triangles τ_1, τ_2 having two vertices v, w in common, which overlap on an open set. Let x_j be the third vertex of τ_j . After relabeling if necessary, we can assume that some initial segment s of the ray $\overrightarrow{vx_1}$ lies in τ_2 . But then let β be the triangle of T' that shares the edge vx_1 with τ_1 . The triangle β only has the vertex v in common with τ_2 but nevertheless intersects τ_2 in a set that contains the segment s . This is a contradiction. ♠

The Disjoint Case: Suppose (Δ_0, Δ_1) is a pair of triangles with no common vertices. Given a vector L we define

$$m_j(L) = \min_{v \in \Delta_j} v \cdot L, \quad M_j(L) = \max_{v \in \Delta_j} v \cdot L. \quad (6)$$

By convexity, we can compute these quantities just by computing the dot products on the vertices. We call (Δ_0, Δ_1) λ -separated if there exists a vector L with $\|L\|_\infty \leq 1$ such that one of the two equations holds:

$$M_0(L) + \lambda < m_1(L), \quad M_1(L) + \lambda < m_0(L), \quad (7)$$

We have $\Delta_0 \cap \Delta_1 = \emptyset$ as long as this pair is 0-separated.

We show by direct calculation that all 24 pairs are 2×10^{-4} separated. Let Q be the set of integer vectors (x, y, z) with $\max(|x|, |y|, |z|) = 300$. We simply enumerate all vectors in the scaled set $\frac{1}{300}Q$ and find that at least one works for each pair. We make this an integer calculation by scaling up these vectors by 300 and scaling up T by 10^{32} . Our separation constant for this scaled problem is 6×10^{30} .

Let T' be another example with $|T' - T| \leq 10^{-4}$. Consider the triangle pair (Δ'_0, Δ'_1) of T' that corresponds to (Δ_0, Δ_1) . In moving from Δ_j to Δ'_j , no vertex moves by more than 10^{-4} , and at most one of the coordinates changes per vertex. Hence, each dot product above changes by at most 10^{-4} . By the triangle inequality, (Δ'_0, Δ'_1) is 0-separated. Hence $\Delta'_0 \cap \Delta'_1 = \emptyset$.

The Common Vertex Case: Now we consider when (Δ_0, Δ_1) have a single common vertex v . Let e_j be the edge of Δ_j opposite v . This time define

$$m_j(L) = \min_{w \in e_j} w \cdot L, \quad M_j(L) = \max_{w \in e_j} w \cdot L. \quad (8)$$

This time we call (Δ_0, Δ_1) λ -separated if there exists a vector L with $\|L\|_\infty \leq 1$ such that one of the two equations holds:

$$M_0(L) + \lambda < v \cdot L < m_1(L) - \lambda, \quad M_1(L) + \lambda < v \cdot L < m_0(L) - \lambda, \quad (9)$$

If (Δ_0, Δ_1) is 0-separated then $\Delta_0 \cap \Delta_1 = \{v\}$. We show by exactly the same kind of calculation that all 72 pairs are 2×10^{-4} -separated.

But now the same argument as above shows that for T' (as above) all such pairs are 0-separated. Hence $\Delta'_0 \cap \Delta'_1 = \{v'\}$.

The two cases above combine with Lemma 3.2 to prove that T' is embedded when $|T' - T| < 10^{-4}$. Hence T is 10^{-4} -robustly embedded.

Remark: There is another approach we could have taken which involves computing the signs of the volumes of all the tetrahedra made from vertices of T . See [DS].

3.2 The Proof Modulo a Crude Bound

Let $G : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be a smooth map. Let $B_r(p)$ denote the ball of radius r about p . We say that G is λ -expansive on $B_r(p)$ if, for all unit vectors V and all $q \in B_r(p)$, we have $\|dG_q(V)\| > 2\lambda$ and $\angle(dF_p(V), dF_q(V)) < \pi/3$.

Lemma 3.3 (Surjectivity) *Let $q = G(p)$. If G is λ -expansive on $B_r(p)$ then $B_{\lambda r}(q) \subset G(B_r(p))$.*

Proof: Let O be the origin and let $B = B_1(O)$. By scaling and translation, it suffices to prove that $B \subset G(B)$ when $p = q = O$ and $\lambda = 2$. Let α be any line segment connecting O to a point on ∂B . Let $s \mapsto \alpha(s)$ be the unit speed parametrization of α . Let $\beta = G(\alpha)$. The parametrization $s \mapsto \beta(s)$ has speed at least 2. Since $\angle(\beta'(t), \beta'(0)) < \pi/3$, the projection of β onto the line through $\beta'(0)$ moves with speed greater than 1. But then $\|\beta(1) - \beta(0)\| > 1$. This shows that $G(\partial B)$ lies entirely outside B . But now the angle condition implies that $G : \partial B \rightarrow \mathbf{R}^3 - B$ is homotopic to the identity map in $\mathbf{R}^3 - B$. Hence $G(\partial B)$ is the generator of the homology group $H_2(\mathbf{R}^3 - q)$ for any $q \in B$. This easily implies that $B \subset G(B)$. ♠

Let $F(z_0, z_2, z_3) = (\theta_0, \theta_2, \theta_3)$ be the map from §2. Let $p \in \mathbf{R}^3$ be the point that corresponds to our torus T from Equation 1. We prove the following result in the next section.

Lemma 3.4 (Crude Bound) *Throughout $B_{10^{-4}}(p)$ we have*

$$\left| \frac{\partial^2 \theta_k}{\partial z_i \partial z_j} \right| < 10^9, \quad \forall i, j, k \in \{1, 2, 3\}.$$

Here is the main corollary.

Lemma 3.5 (Expansion) *F is 10^{-2} -expansive on $B_{10^{-12}}(p)$.*

Proof: Let V be a unit vector. Define

$$M = \begin{bmatrix} -1.66 & +1.46 & -0.02 \\ +1.46 & +0.04 & -1.13 \\ -0.02 & -1.13 & +1.91 \end{bmatrix} \quad (10)$$

M is a symmetric and its eigenvalues all exceed $3/10$ in absolute value. Hence $\|M(V)\| > 3/10$. We will settle for the weaker bound:

$$\|M(V)\| > \frac{1}{10}. \quad (11)$$

Define “error matrices” E_q and E'_q so that

$$dF_q = M + E_q = dF_p + E'_q, \quad E_q = E_p + E'_q. \quad (12)$$

Comparing Equations 3 and 10 we see that

$$\|E_p\|_\infty < \frac{1}{100}. \quad (13)$$

We can join p to q by an axis-aligned path of length at most $\sqrt{3} \times 10^{-12}$. Hence, by the Crude Bound Lemma and integration:

$$\|E'_q\|_\infty < \sqrt{3} \times 10^{-12} \times 10^9 < \frac{1}{450} \implies \|E'_q(V)\| < \frac{1}{150}. \quad (14)$$

By Equation 13, Equation 14, and the triangle inequality,

$$\|E_q\|_\infty \leq \|E_p\|_\infty + \|E'_q\|_\infty < \frac{1}{100} + \frac{1}{450} < \frac{1}{75} \implies \|E_q(V)\| < \frac{1}{25}. \quad (15)$$

By Equation 11, Equation 15, and the triangle inequality,

$$\|dF_q(V)\| \geq \|M(V)\| - \|E_q(V)\| > \frac{1}{10} - \frac{1}{25} > 2 \times \frac{1}{100}. \quad (16)$$

By Equation 14, Equation 16, and some elementary geometry,

$$\frac{\|E'_q(V)\|}{\|dF_p(V)\|} < \frac{1/150}{1/50} = \frac{1}{3} < \frac{\sqrt{3}}{2} \implies \angle(dF_p(V), dF_q(V)) < \frac{\pi}{3}. \quad (17)$$

Equations 16 and 17 give the desired bounds. ♠

Let $q = F(p)$ and let $q^* = (2\pi, 2\pi, 2\pi)$. We compute directly, in Mathematica, that $\|q - q^*\| < 10^{-31}$. By the Surjectivity and Expansion Lemmas, $q^* = F(p^*)$ with $\|p - p^*\| < 10^{-29}$. The torus T^* corresponding to p^* is flat. Since T is 10^{-4} robustly embedded, we see that T^* is embedded. The dihedral angles of T all lie in $[\cdot 005, \pi - \cdot 005]$, so the tiny change we make in going from T to T^* keeps T^* a pup tent. This completes the proof of Theorem 1.2 modulo the Crude Bound Lemma.

Remark: All we used for our proof is that T is 10^{-12} robustly embedded and that $\|q - q^*\| < 10^{-14}$. One can get these weaker bounds from ordinary floating point operations in Java.

3.3 Proof of the Crude Bound Lemma

We keep the notation from above. Note that when we vary (z_1, z_2, z_3) , we also vary (z_6, z_5, z_4) in a symmetric way. Thus,

$$\frac{\partial \theta_j}{\partial z_1} = \partial_1 \theta_j + \partial_6 \theta_j, \quad \frac{\partial^2 \theta_j}{\partial z_1 \partial z_2} = (\partial_{12} + \partial_{62} + \partial_{15} + \partial_{65}) \theta_j.$$

$\partial_i \theta_j$ denotes the rate of change of θ_j when we vary *only* z_i and $\partial_{ij} \theta_k$ denotes the rate of change of $\partial_j \theta_k$ when we vary only z_i . Here we are leaving the realm of 2-fold symmetric configurations and just considering 8 points in space.

Each θ_j is a 6-term expression of terms of the form θ_{ijk} , where θ_{ijk} is the angle of triangle (i, j, k) at vertex i . Hence, $\partial_{ij} \theta_j$ is a 24-term sum of expressions of the form $\partial_{ab} \theta_{cde}$. These expressions vanish when $\{a, b\} \not\subset \{c, d, e\}$. We call this expression *friendly* if $c \notin \{a, b\}$, and otherwise *unfriendly*.

Lemma 3.6 *Every unfriendly term is a sum of at most 2 friendly terms.*

Proof: Consider terms of $\partial_{aa} \theta_{abc}$. Using the relation $\theta_a + \theta_b + \theta_c = \pi$ we have

$$\partial_{aa} \theta_{abc} + \partial_{aa} \theta_{bca} + \partial_{aa} \theta_{cab} = 0.$$

The terms on the right are friendly.

Given that $\partial_{ab} = \partial_{ba}$, the only remaining unfriendly terms we have to treat are those of the form $\partial_{ba} \theta_{abc}$ and $\partial_{ba} \theta_{acb}$. We will treat the first one. The second one has the same treatment.

We also have the relation $(\partial_a + \partial_b + \partial_c) \theta_{abc} = 0$. Geometrically, the variation we are considering simply translates the triangle up or down along the Z -axis and changes none of the angles. Applying ∂_b to this relation, we have

$$\partial_{ba} \theta_{abc} = -\partial_{bb} \theta_{abc} - \partial_{bc} \theta_{abc}.$$

Both terms on the right are friendly. ♠

In summary, each partial derivative in the Crude Bound Lemma is a sum at most 48 friendly term. To prove the Crude Bound Lemma it suffices to show that each friendly term is less than 10^7 in absolute value throughout the ball $B_{10^{-4}}(p)$.

We say that a torus is *relevant* if it corresponds to a point in our tiny ball B . We say that a pair (V_1, V_2) of vectors is *relevant* if it is obtained from a triangle (P_0, P_1, P_2) of a relevant torus by setting $V_1 = P_1 - P_0$ and $V_2 = P_2 - P_0$. Each triangle in a relevant torus gives rise to 6 pairs of relevant vectors.

Let $\vartheta(V_1, V_2)$ be the angle between V_1 and V_2 . Each friendly term has the form

$$\frac{\partial^2 \vartheta}{\partial w_i \partial w_j}, \quad \vartheta = \arccos \left(\frac{V_1 \cdot V_2}{\sqrt{(V_1 \cdot V_1)(V_2 \cdot V_2)}} \right) \quad (18)$$

The square g_{ij} of the expression on the left side of Equation 18 is a rational function. To prove the Crude Bound Lemma, we just need to prove for all relevant pairs and all (i, j) that that $g_{ij} < 10^{14}$.

Lemma 3.7 *For each relevant pair (V_1, V_2) of vectors,*

$$\|V_1 \times V_2\|^2, \|V_1\|^2, \|V_2\|^2 \in (10^{-1}, 10).$$

Proof: For T we get the bounds $\|V_k\| \in [0.48, 2.4]$ and $\|V_1 \times V_2\| \in [.85, 2.4]$. These various quantities hardly change at all when we switch to a different relevant torus and change our coordinates by less than 10^{-4} . ♠

Case 1: Let $W_k = (u_k, v_k, 0)$ be the projection of V_k to the XY -plane. We have

$$g_{21} = g_{12} = \frac{\|W_1 \times W_2\|^4}{\|V_1 \times V_2\|^6} \leq \frac{\|V_1 \times V_2\|^4}{\|V_1 \times V_2\|^6} = \frac{1}{\|V_1 \times V_2\|^2} < 10.$$

The second inequality comes from Lemma 3.7.

Case 2: With W_k defined as above, we have

$$g_{11} = \frac{\left(t_1 w_1 w_2 + t_2 w_1 w_2^3 + t_3 w_1^3 w_2 + t_4 w_1^2 w_2^2 + t_5 w_1^4 + t_6 w_1^2 + t_7 \right)^2}{\|V_1\|^8 \|V_1 \times V_2\|^6},$$

$$\begin{aligned} t_1 &= 3 \|W_1\|^2 \|W_1 \times W_2\|^2, & t_2 &= 2 \|W_1\|^4, \\ t_3 &= 3 (W_1 \cdot W_2)^2 + 3 \|W_1\|^2 \|W_2\|^2, & t_4 &= -6 \|W_1\|^2 (W_1 \cdot W_2), \\ t_5 &= -2 \|W_2\|^2 (W_1 \cdot W_2), & t_6 &= -\|W_1 \times W_2\|^2 (W_1 \cdot W_2), \\ t_7 &= \|W_1\|^2 \|W_1 \times W_2\|^2 (W_1 \cdot W_2). \end{aligned}$$

Write this as $g_{11} = N^2/D$. Using Lemma 3.7 and the fact that $|w_1|, |w_2| < 1$ we see that $|N| < 10^{7/2}$ and $D > 10^{-7}$. Hence $g_{11} < 10^{14}$. A similar calculation works for g_{22} and indeed this case follows from that of g_{11} and symmetry.

This completes the proof of the Crude Bound Lemma, and thereby completes the proof of Theorem 1.2. An 8-vertex paper torus exists!

4 Discussion and Pictures

4.1 Experimentation

Recall that there are 7 combinatorial types of triangulation of the torus having 8 vertices. Figure 1.2 shows the best of these, the one with uniform degree. For all but the worst triangulation, I generated lots of embedded examples by randomly sampling collections of 8 points on the unit sphere and testing the embedding condition.

Next, I used a hill climbing algorithm. The way this works is that you want to minimize an objective function on a space. You start at a random point and then evaluate the function. You then make a random perturbation and move to the new location if the function is lower. If you are in a smooth manifold and the function is smooth and nonsingular, then the random perturbations often move in the direction of the gradient. In this setting the hill climbing algorithm somewhat mimics gradient flow. This approach works pretty well in low dimensions.

My hill-climbing algorithm started with random embedded examples on the unit sphere and repeatedly perturbed them – not necessarily keeping them on the unit sphere – with the goal of minimizing an objective *flatness function*. The flatness function computes the maximum deviation of a cone angle from being 2π .

After a lot of experimentation, I implemented 4 tricks that helped immensely.

1. It was useful to first perturb the points within an ellipsoid surface that contains them, and then to perturb the ellipsoid. This method is guaranteed to keep all vertices on the convex hull boundary.
2. It was extremely useful to control the combinatorics on the convex hull boundary. The convex hull boundary has 12 faces, some of which – say K of them – are also faces in the triangulation of the torus. Call K the *face number*. I noticed that the examples with large face number never led to flat examples. So I eventually constrained my search to examples with low face number.
3. It helped to bound some auxiliary quantities, like the minimum angles and dihedral angles, to prevent degenerations. Bounds of about 0.00001 on the dihedral angles worked well.
4. It helped to force the examples to have 2-fold rotational symmetry. This simplification came later. After I found some asymmetric examples, Samuel Lelièvre asked me about symmetric examples and I tweaked the program. The program works more reliably after incorporating the symmetry.

4.2 Pictorial Study

With all the tricks above in place, and a lot of playing around, I found some embedded and 2-fold symmetric examples of face number 6 which are flat up to 10^{-16} . I took the nicest one and simplified the coordinates, feeding it back into the program to make it near-flat again. After some fooling around with the digits I arrived at the example in Equation 1.

Figure 4.1 shows three Mathematica plots of the pup tent, from various angles.

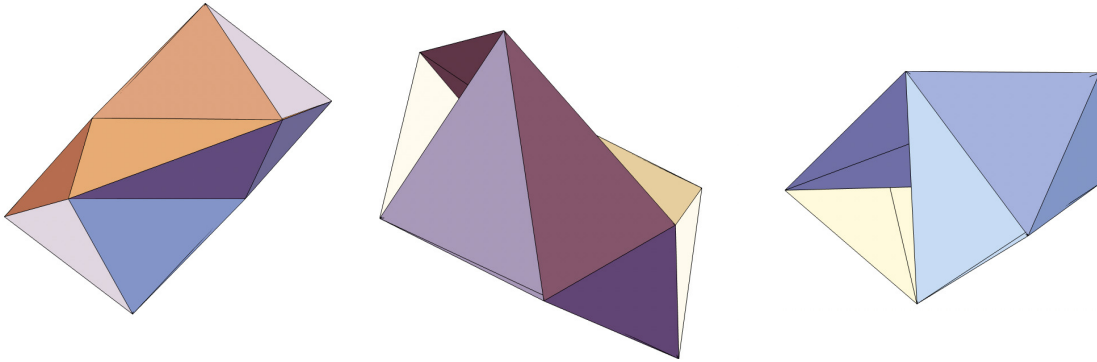


Figure 4.1: 3D plots of the pup tent

Figure 4.2 shows a numerical computation of the intrinsic geometric structure of the faces of the pup tent. We are showing part of the universal cover of the triangulation. We have also highlighted the fundamental domain that corresponds to the one in Figure 1.2. The blue triangles correspond to the faces of the pup tent on the convex hull boundary. There are 6 of them in the fundamental domain.

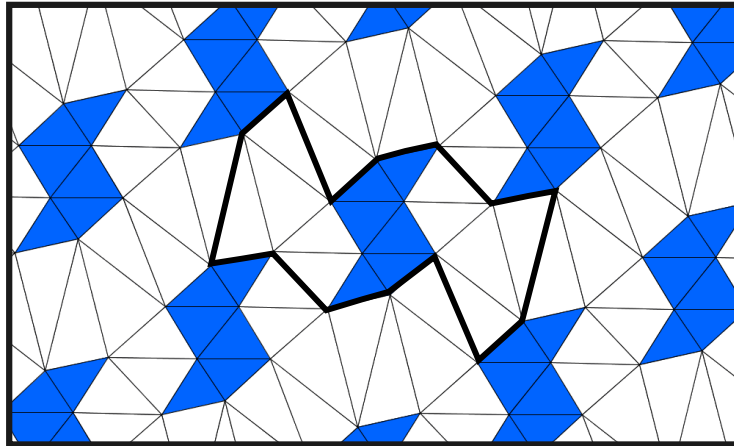


Figure 4.2: The intrinsic structure.

Figure 4.3 shows a second coloring of the same fundamental domain. We have labeled the points. We list out numerical approximations to the 8 blue triangles. We get the pink ones by reflecting the blue ones in the origin and applying the permutation $j \rightarrow 7 - j$ to the labels.

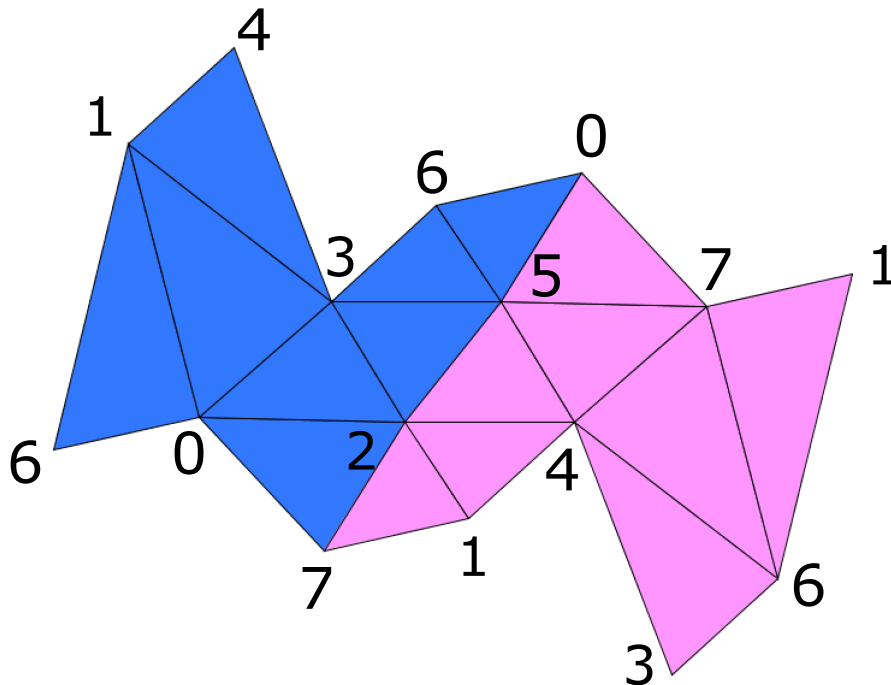


Figure 4.3: A fundamental domain for the tiling

$\{6, 0, 1\}$: $\{-2.9340654093 - 0.6455398803i, -1.8673131665 - 0.4088246583i, -2.3837468425 + 1.5965906856i\}$,
 $\{3, 1, 0\}$: $\{-0.8925492572 + 0.4429076765i, -2.3837468425 + 1.5965906856i, -1.8673131665 - 0.4088246583i\}$,
 $\{3, 4, 1\}$: $\{-0.8925492572 + 0.4429076765i, -1.6100587412 + 2.3020419213i, -2.3837468425 + 1.5965906856i\}$,
 $\{2, 0, 7\}$: $\{-0.3555738883 - 0.4429076765i, -1.8673131665 - 0.4088246583i, -0.9478910867 - 1.3850741341i\}$,
 $\{3, 0, 2\}$: $\{-0.8925492572 + 0.4429076765i, -1.8673131665 - 0.4088246583i, -0.3555738883 - 0.4429076765i\}$,
 $\{3, 2, 5\}$: $\{-0.8925492572 + 0.4429076765i, -0.3555738883 - 0.4429076765i, +0.3555738883 + 0.4429076765i\}$,
 $\{3, 5, 6\}$: $\{-0.8925492572 + 0.4429076765i, +0.3555738883 + 0.4429076765i, -0.1188611560 + 1.1483589122i\}$,
 $\{5, 0, 6\}$: $\{+0.3555738883 + 0.4429076765i + 1.8673131665 + 0.4088246583i, +0.9478910867 + 1.3850741341i\}$

Figures 4.2 and 4.3 look quite tame, but the extrinsic geometry of the pup tent is beautiful and very intricate. Figure 4.4 shows one of the slices of the pup tent by a plane parallel to the Z axis. The solid handlebody bounded by the torus is colored orange. This slice is very spiky and sort of “barely embedded”, but the embeddedness is robust compared to the plot precision.

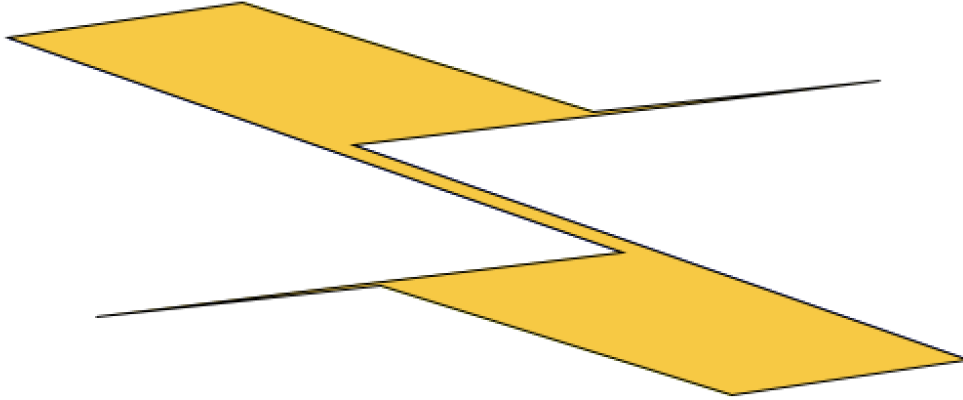


Figure 4.4: A horizontal slice of the pup tent

Figure 4.5 shows the slice by the XZ plane, which goes through the axis of symmetry of the torus. The slice of the handlebody is an annulus. We have included a closeup of the bottom so that the reader can see it better.

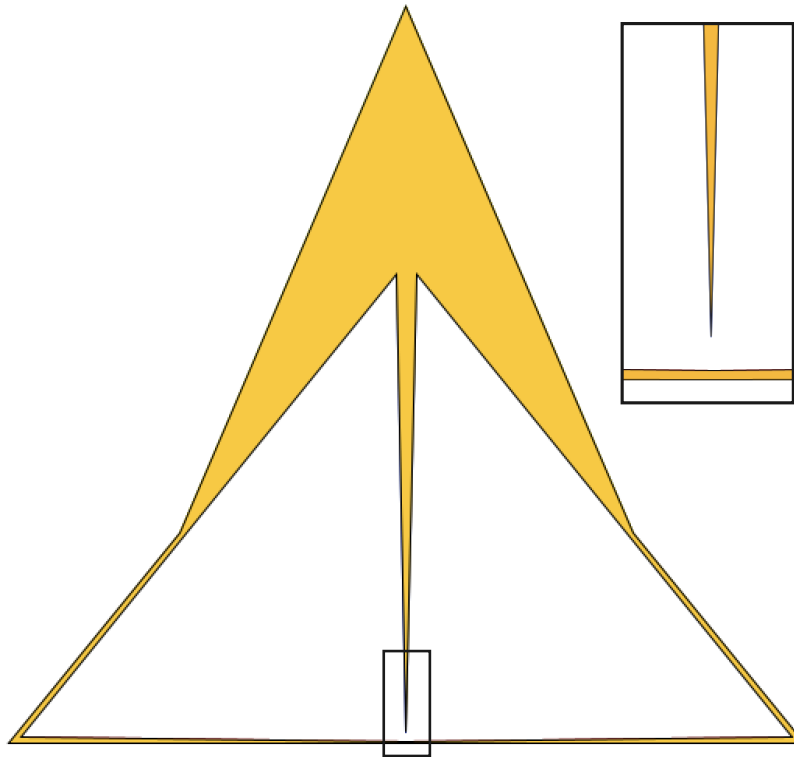


Figure 4.5: The XZ planar slice of the pup tent.

Figure 4.6 shows the slice by the YZ plane, with a closeup of the bottom.

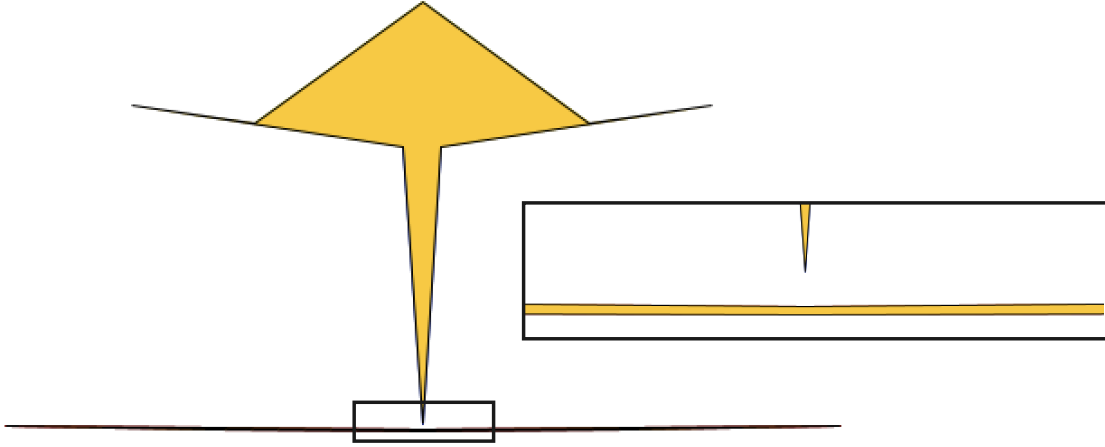


Figure 4.6: The YZ planar slice of the pup tent

The left side of Figure 4.7 shows a random slice of the pup tent.

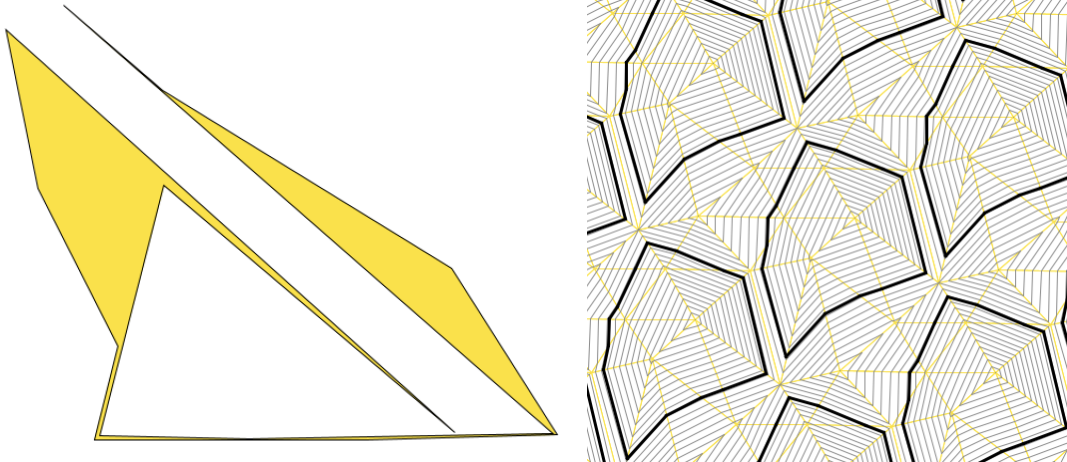


Figure 4.7: Extrinsic and Intrinsic structures compared

Once we fix this direction we get a *level set foliation* of the pup tent by curves that lie in planes perpendicular to this direction. On the right side of Figure 4.7, we have drawn the level set foliation in the intrinsic flat structure and then lifted the picture to the universal cover. The highlighted loop on the right corresponds to the loop shown on the left.

My supplementary notes [S] give additional information about the pup tent, including a pattern than you can cut out and (with effort) fold into one. Also, my Java program lets you see many more pictures like this.

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